

Introduction to Computer Architecture

Sheet #2

Exercises:

1. The following memory units are specified by the number of words times the number of bits per word. How many address lines and input-output data lines are needed in each case?
 - a) 2K x 16
 - b) 64K x 8
 - c) 16M x 32
 - d) 96K x 12.
2. If a ROM has a total capacity of 32K bits and 4 outputs, how many address lines are there?
3. How many bits would you need to address a $2\text{M} \times 32$ memory if:
 - a) The memory is byte-addressable?
 - b)** The memory is word-addressable.
4. Suppose that a $2\text{M} \times 16$ main memory is built using $256\text{KB} \times 8$ RAM chips and memory is word-addressable.
 - a) How many RAM chips are necessary?
 - b) How many RAM chips are there per memory word?
 - c) How many address bits are needed for each RAM chip?
 - d) How many banks will this memory have?
 - e) How many address bits are needed for all of memory?
 - f) If high-order interleaving is used, where would address 14 (which is E in hex) be located?
 - g) Repeat Exercise (f) for low-order interleaving.
5. A digital computer has a memory unit with 24 bits per word. The instruction set consists of 150 different operations. All instructions have an operation code part (opcode) and an address part (allowing for only one address). Each instruction is stored in one word of memory.
 - a) How many bits are needed for the opcode?
 - b) How many bits are left for the address part of the instruction?
 - c) What is the maximum allowable size for memory?
6. Assume a 2^{20} byte memory:
 - a) What are the lowest and highest addresses if memory is byte-addressable?
 - b) What are the lowest and highest addresses if memory is word-addressable, assuming a 16-bit word?